

# Asjad Yousaf Khan

## Junior AI/ML Engineer

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### Professional Summary

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Junior AI/ML Engineer and Computer Science graduate with hands-on experience in machine learning, deep learning, computer vision, natural language processing, retrieval-augmented generation, LangChain, vector databases, and full-stack AI applications. Skilled in Python, data preprocessing, feature engineering, model training, model evaluation, image processing, semantic search, embeddings, and AI chatbot development. Built practical projects involving object detection, face detection, speech-based interaction, document question answering, image restoration, sentiment analysis, disease prediction, plagiarism detection, and deepfake detection.

### Technical Skills

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**Programming Languages:** Python, SQL, JavaScript, C++

**Machine Learning:** Supervised Learning, Classification, Regression, Data Preprocessing, Feature Engineering, Model Training, Model Evaluation

**Deep Learning and Computer Vision:** Object Detection, Face Detection, Image Processing, Image Restoration, Visual Classification, Deepfake Detection

**NLP and Generative AI:** Natural Language Processing, Sentiment Analysis, RAG, LangChain, Vector Databases, Embeddings, Semantic Search, Prompt Engineering

**Libraries and Tools:** Scikit-learn, Pandas, NumPy, TensorFlow, PyTorch, OpenCV, Jupyter Notebook, Google Colab, Git, GitHub, VS Code

**Software Development:** REST APIs, Full-Stack Development, SQL Databases, E-commerce Applications, Inventory Management Systems

### Education

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**Bachelor's Degree in Computer Science**

**Quaid-i-Azam University, Islamabad**

Relevant Coursework: Machine Learning, Artificial Intelligence, Deep Learning, Computer Vision, Natural Language Processing, Data Structures and Algorithms, Database Systems, Software Engineering

### Certifications

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**Google Certified Professional Machine Learning Engineer**

*Udemy Certificate*

### Projects

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**AI Vision Assistant App – Final Year Project** *Computer Vision, Speech Input/Output, Raspberry Pi, Mobile AI*

- Built an assistive vision application with mobile and Raspberry Pi modes for camera-based environment understanding.
- Implemented object detection, face detection, voice commands, and audio feedback to support hands-free interaction.
- Designed an end-to-end AI workflow that combines image input, visual recognition, speech input, and spoken output.

**V Prep – AI Interview Preparation Platform**

*LangChain, NLP, Generative AI, STAR Method*

- Developed an AI interview preparation platform that generates interview questions, contextual follow-ups, and practice sessions.
- Used LangChain and NLP workflows to support dynamic questioning, structured responses, and personalized feedback.
- Applied the STAR method to help users organize behavioral interview answers clearly and professionally.

**CS QAU Chatbot**

*RAG, Vector Database, Embeddings, Semantic Search*

- Created a retrieval-augmented chatbot for department-related document question answering.
- Implemented embeddings, vector search, semantic retrieval, and conversational response generation.
- Improved information access by retrieving relevant context from stored documents before generating answers.

**Pixify – Image Enhancement and Restoration System**

*Computer Vision, Image Processing*

- Built an image restoration system for old, damaged, torn, and low-quality photos using image processing techniques.
- Applied preprocessing, enhancement, noise reduction, and detail restoration workflows to improve image clarity.

**Deepfake Detection using FaceForensics++**

*Deep Learning, Computer Vision, Classification*

- Developed a deepfake detection pipeline using the FaceForensics++ dataset for manipulated media classification.
- Performed preprocessing, feature analysis, model training, and evaluation for visual authenticity detection.

### Additional Projects

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**Disease Predictor:** Machine learning prediction model using structured health-related features.

**Opinion Miner:** NLP sentiment analysis system for opinion mining and text classification.

**Machine Learning Practice Portfolio:** 15 machine learning projects covering classification, regression, preprocessing, training, and evaluation.

**AI Recipe Bot:** Conversational AI assistant for recipe suggestions based on user input.

**AI Plagiarism Checker:** NLP and text similarity system for plagiarism detection.

**Web Development Projects:** SpareKart.live, Dine3D, Kitchub.store, Tamkeenzone.pk, Ebnalarab.com.